

Sprint Update

Week 2 — 27 August to 4 September

Eighteen executed experiments re-tested the lower-back-first research question, probed data enrichment, synthetic ge

1

Model improvement
admitted

11+

Enrichment / rescue
experiments rejected

34

Notebooks executed
to date

114.75 GiB

Redundant storage
reclaimed

Source: [wiki/log.md](#) (2026-09-01 to 2026-09-03) · [reports/*_2026-09-0{1,2,3}.md](#)

What Changed This Sprint

0.8882

New incumbent AUROC
Lower-back ensemble

0.757

Lower-back-only external
Failed RevalExo gate

52.2%

Non-stroke false-positive rate
138 hard negatives, 0.50 cutoff

93.1%

NONAN healthy specificity
29 frozen participants



What was admitted

- A fixed equal-probability **ERM + Deep CORAL + ERM++ ensemble** on lower-back acceleration — passed every non-inferiority and material-gain gate
- A rigorous **test-set credibility audit** quantifying exactly how much more paired data is required
- A formal **cross-dataset evidence gate** that now governs every future enrichment attempt



What was rejected

- Lower-back-only as a replacement prototype (failed external gate)
- MAREA/DUO-GAIT/NONAN healthy-pool enrichment (domain-shift false positives)
- Six synthetic-data generators (interpolation, recombination, VAE ×2, diffusion)
- Signed-axis representation, pooled gyroscope, MiniROCKET fusion, MIL, canonical architecture swaps

Source: reports/MODEL_IMPROVEMENT_GATE_2026-09-03.md ·

reports/LOWER_BACK_ONLY_EXTERNAL_BENCHMARK.md ·

reports/VOISARD_NONSTROKE_HARD_NEGATIVE_EVALUATION_2026-09-01.md · wiki/log.md

The Lower-Back-First Research Question

A single lumbar sensor is the more deployable prototype. It was tested as a candidate primary model under the same folds as the 3-channel comparator.

Internal 5-Fold (matched protocol)

Model	AUROC	Brier	Bal. Acc.
3-channel LB/LF/RF	0.966	0.074	0.926
Lower-back only	0.933	0.121	0.829

Lower-back-only trails by 0.033 AUROC and 0.096 balanced accuracy — a credible single-sensor baseline, not yet equivalent.

Frozen RevalExo Gate (n=17)

Model	AUROC	Brier	Bal. Acc.
3-channel comparator	0.914	0.161	0.714
Lower-back only	0.757	0.217	0.571

❗ Lower-back-only fails the external robustness gate. Decision: retain it as a transparent single-sensor comparator, not the final prototype.

Source: reports/LOWER_BACK_VS_THREE_CHANNEL_BENCHMARK.md ·
reports/LOWER_BACK_ONLY_EXTERNAL_BENCHMARK.md ·
reports/LOWER_BACK_CHANNEL_FAILURE_ANALYSIS.md

Healthy-Pool Enrichment — Rejected

Tier-1 MAREA + DUO-GAIT Enrichment

5,938 adapted healthy windows from 36 participants (MAREA 3,526 · DUO-GAIT 2,412) were added to training, matched units and resampling verified first.

Frozen RevalExo	Baseline	Enriched
AUROC	0.914	0.929 ↑
Brier	0.161	0.177 ↓
Balanced accuracy	0.714	0.571 ↓
Healthy FP	4/7	6/7 ↓

Root-Cause Diagnosis

MAREA lower-back magnitude (1.059) is measurably higher than every existing healthy source, despite verified unit conversion and resampling.

Source	LB mean	LF mean	RF mean
Felius	1.023	1.740	1.790
Voisard	1.030	1.535	1.586
DUO-GAIT	1.032	1.649	1.704
MAREA	1.059	1.615	1.618

Verdict: residual domain shift + probability miscalibration, not "enrichment is useless." Not re-attempted without source-conditional calibration.

Synthetic Healthy-Data Generation — All Candidates Rejected

Every generator remained training-excluded pending realism, memorisation, and downstream frozen-external gates. None reached admission.

Method	Source	Quality result	Downstream ablation	Verdict
Convex phase-mixing	MAREA	Std. dev. 0.661 vs 0.882 real; spectral energy 0.0058 vs 0.0139 real	Not run — failed realism first	Rejected
Cross-fade recombination	MAREA	Passed structural checks	RevalExo Brier 0.161→0.172, Bal. Acc. 0.714→0.643	Rejected
Convolutional VAE (v1)	MAREA cycles	Oversmoothed; not admitted	Not run	Rejected

NONAN GaitPrint — Large-Scale Healthy Specificity Test

29

Frozen participants
23,657 windows

93.1%

Healthy specificity
Wilson CI 78.0–98.1%

80

Candidate participants
68,398 windows

Not admitted

Repeated paired gate
No durable benefit

Frozen score (reporting-only)

At the fixed 0.50 reference, 2/29 healthy participants were misclassified (specificity 93.1%). The two false positives share no age group, mobility-screen flag, or amplitude profile — ruling out a simplistic clipping/scaling explanation.

Candidate enrichment — rejected

A bounded, capped, source-downweighted candidate addition looked promising in one pass (candidate specificity 0.9107→0.9630), but the **repeated 15-unit paired gate** found no durable benefit: AUROC Δ +0.0002 (CI -0.0039 to +0.0049), specificity Δ -0.0166 (CI -0.0901 to +0.0480). A class-aware CORAL alignment variant also failed.

Source: wiki/log.md (2026-09-02) · notebooks/14–18_nonan_*.ipynb

Representation Audits — Magnitude Confirmed as the Deployment Contract

Signed Axes vs. Magnitude

Representation	Internal AUROC	RevalExo AUROC	Healthy FP
Magnitude (3ch)	0.979	0.914	4/7
Signed axes (9ch)	0.974	0.457	7/7

Signed axes look competitive internally but collapse externally — not orientation/domain invariant across source hardware. Magnitude remains the official contract.

Lower-Back Gyroscope (6-DoF)

Held-out source	Accel-only	+Gyroscope
Felius (transport gain)	0.850	0.859
Sint (transport loss)	0.870	0.772
Sint healthy specificity	0.750	0.433

Gyroscope magnitude verified via a new Sint 6-DoF adapter (4,053 windows). Rejected as a pooled replacement — helps one transport direction, materially hurts another.

Source: reports/PIPELINE_RESEARCH_COMPARISON_AND_PROCESSING_AUDIT_2026-09-01.md · wiki/log.md (2026-09-02, lower-back 6-DoF)

Normalization & Augmentation Probes — Baseline Retained

Global Z-Score vs. Robust Median/IQR

Track	Norm.	AUROC	Bal.Acc	Healthy FP
Lower back	Z-score	0.767 $\pm$.058	0.591	5.53
Lower back	Robust	0.772 $\pm$.032	0.591	5.53
3-channel	Z-score	0.917	0.661	4.33

Nuisance-Augmentation Probe

Per-window gain ($\pm 10\%$), Gaussian noise, and ± 8 -sample temporal roll during training only.

Track	AUROC	Brier	Healthy FP
Lower-back (aug.)	0.751	0.224	6.2
3-channel (aug.)	0.917	0.163	5.0

⊘ Degraded external calibration and specificity — rejected outright.

The Critical Gap: Non-Stroke Hard Negatives

Same-protocol Voisard cohorts with other neurological/orthopedic conditions — never relabelled, never pooled into training.

Cohort	N	Meaning	FP @ 0.50
ACL	11	Orthopedic	36.4%
CIPN	19	Neurological	52.6%
HOA	15	Orthopedic	26.7%
KOA	18	Orthopedic	38.9%
PD	24	Neurological	62.5%
RIL	51	Neurological	62.7%
Pooled	138	Non-stroke only	52.2%

Not a stroke-specific classifier

Parkinson's disease and radiation-induced leukoencephalopathy score stroke-like most often. Raising the threshold to 0.78 reduces pooled FP to 31.2% but also reduces stroke sensitivity.

Binary exposure experiment — also rejected

Adding a temporary hard-negative exposure target during training either damaged primary calibration (25% dose: Brier 0.090→0.140) or gave an insufficient specificity gain (7.7% dose: FP only 52.2%→46.4%). Neither replaces the binary baseline.

Cross-Dataset Evidence Gate — Research Discipline Formalised

Before proposing another modelling approach, eight literature/code bases (BenchHAR, HAROOD, ContrastSense, CALDA/CALDG, DAGHAR, IMUDiffusion, PPDA/WIMUSim, stroke/lower-back studies) were audited against this project's exact task, sensors, and split rules.

Why past negative results don't contradict the literature

- Many reported successes concern generic HAR or gait detection, not disease classification
- Many use target-domain data during fitting — this project explicitly forbids that
- Accelerometer-only can outperform accelerometer+gyroscope — consistent with this project's own finding
- Synthetic gains can reverse for individual held-out people or larger configurations

Mandatory checklist for every future experiment

- Official-code trace before re-implementing any method
- Matched ERM baseline under the same protocol
- Participant/source-disjoint validation, always
- Training-only generator fitting; real-only validation/testing
- Worst-source clinical metrics reported, not just pooled means
- No RevalExo/NONAN used for model selection — ever

Source: [reports/EVIDENCE_GATE_CROSS_DATASET_IMU_2026-09-03.md](#)

Admitted: The Lower-Back ERM + CORAL + ERM++ Ensemble

0.8882

AUROC

was 0.8737

0.1425

Brier

was 0.1664

0.8140

Bal. accuracy

was 0.7834

0.7743

Specificity

was 0.7145

0.8537

Sensitivity

was 0.8523

Protocol

- 314 participants, 22,506 lower-back windows, Felius + Voisard + Sint
- Leave-one-complete-source-out outer evaluation, 5 seeds
- Fixed equal-probability average of ERM, Deep CORAL, ERM++-style head warm-up
- RevalExo and NONAN **not loaded** for training, tuning, or selection

Paired bootstrap (ensemble – ERM)

Metric	Δ mean	95% CI
Balanced accuracy	+0.0306	+0.004 to +0.064
Specificity	+0.0599	−0.001 to +0.132
AUROC	+0.0146	+0.001 to +0.032
Brier	−0.0239	−0.049 to −0.002

Rescue Attempts After the Ensemble — All Rejected

Every candidate had to beat the incumbent ensemble (FP 9.80, FN 11.93, 21.73 total errors per source/seed) on the same three-source, five-seed, leakage-safe protocol.

Candidate	AUROC	Bal.Acc	FP	FN	Total errors	Verdict
Incumbent ensemble	0.8882	0.8140	9.80	11.93	21.73	Selected
+ MiniROCKET fusion (50:50)	0.8869	0.8145	9.33	12.80	22.13	Rejected
Canonical 10k MiniROCKET	0.8572	0.8029	9.40	13.87	23.27	Rejected
Canonical InceptionTime	0.8273	0.7418	13.07	19.07	32.13	Rejected
Participant mean-pooling MIL	0.8442	0.7691	9.80	11.53	21.33	Rejected
Participant gated-attention MIL	0.8043	0.7528	11.00	12.33	23.33	Rejected

Test-Set Credibility Audit — Why More Data Is the Limiting Factor

Seed-Consensus Confusion (development)

	Pred: Healthy	Pred: Stroke
True: Healthy	TN: 99	FP: 27
True: Stroke	FN: 33	TP: 155

Pooled specificity 78.6% (CI 70.6–84.8%) · sensitivity 82.4% (CI 76.4–87.2%). Diagnostic summary, not an independent test — every source influenced selection.

Planning target

At current development rates, a two-sided 95% Wilson interval no wider than 10 points needs approximately **257 independent healthy** and **222 independent stroke** participants — illustrating why 7 healthy / 10 stroke (RevalExo) cannot be a reliability claim.

Required final-test contract (7 rules)

Freeze everything before access · new untouched multi-site participants · participant is the statistical unit · represent age/sex/severity/speed/aids/non-stroke confounders · independent clinical reference diagnosis · full 2×2 + CI reporting · no post-hoc model or threshold update.

Source: reports/TEST_SET_CREDIBILITY_AUDIT_2026-09-03.md · notebooks/30_fp_fn_and_test_set_credibility_audit.ipynb

New External Evidence: Specificity Wins and Candidate Sources

Component Robustness Results

Carpinella 6MWT healthy cohort — 60 healthy participants, 6,109 windows.

Lower-back-only baseline made **0 stroke predictions** at 0.50 (Wilson upper bound 6.0%).

Zenodo stroke-only cohort — 3-channel detected 9/10 stroke (95% CI 59.6–98.2%); lower-back-only detected 10/10 (95% CI 72.2–100%).

Triaxial older-healthy cohort — 53/59 (89.8%) false positives. Diagnosed as a signal-domain/device mismatch (median magnitude 0.578g vs the model's fitted 1.016g), **not** an age effect — must not be pooled or reinterpreted as age bias.

Screened Candidate Sources (not yet acquired for training)

Source	Composition	Role
WearGait-PD	85 healthy + 100 PD, full- body IMU	Non-stroke hard- negative test
6MWT normative	60 healthy, lower-back only	Healthy age/domain reference
BLISS	21 mixed impairment,	Foot/impairment stress test

Housekeeping: Data Storage Retention Audit

268.68 GiB

Before cleanup

153.93 GiB

After Tier-A cleanup

114.75 GiB

Reclaimed

Removed (Tier A — executed)

- 80 candidate-NONAN source archives (107.32 GiB) — enrichment already rejected
- 3 structural-audit NONAN archives (4.07 GiB) — already materialized
- Redundant Sint release ZIP (2.41 GiB) — extraction already verified
- Redundant Mobilise-D ZIP copies (0.95 GiB) — already extracted

Protected — verified intact after cleanup

- Felius, Voisard, extracted Sint, RevalExo (all active sources)
- `data/processed/` — every tensor, metric, and checkpoint
- NONAN compact processed tensors and manifests
- All executed notebooks, reports, and wiki provenance

Source: reports/DATA_STORAGE_RETENTION_AUDIT_2026-09-03.md

Updated Synthesis After the Sprint

NEW INCUMBENT

0.8882

AUROC · lower-back ensemble
Development-selected, 314 participants

3-CHANNEL COMPARATOR UNCHANGED

0.914

AUROC · frozen RevalExo (n=17)
Still the higher-performance track

LIMITING FACTOR

Cohort size

Not architecture — 11 rescue attempts all failed the same gate

What this sprint actually proved

Eighteen executed, leakage-safe experiments converge on one conclusion: **the remaining error is explained by score overlap on a 314-participant development pool, not by model architecture, normalization, representation, or missing healthy-domain data.** Every enrichment, synthesis, representation, and rescue attempt failed the same predefined non-inferiority gate. The lower-back ensemble is a genuine, evidence-gated improvement over plain ERM, but it is **development-selected, not externally validated**, and the 3-channel prototype remains the stronger externally-tested candidate.

Source: reports/MODEL_IMPROVEMENT_GATE_2026-09-03.md ·
reports/CANONICAL_CORRECTIVE_BENCHMARK_2026-09-03.md ·
reports/TEST_SET_CREDIBILITY_AUDIT_2026-09-03.md

Updated Recommended Next Steps

① FREEZE & PACKAGE

Lock the exact lower-back ensemble recipe (members, weights, threshold) — no further tuning on these 314 participants.

② ONE FROZEN EXTERNAL PASS

Evaluate the packaged ensemble on RevalExo exactly once, under a pre-registered protocol, without retuning.

③ RECRUIT NEW PAIRED COHORT

Target ~257 healthy / ~222 stroke participants for a precise sensitivity/specificity interval — the single limiting requirement identified this sprint.

④ NON-STROKE SPECIFICITY

Acquire WearGait-PD or equivalent to test the 52.2% non-stroke false-positive rate against a genuine external cohort.

Closed for now

- Architecture rotation on the existing 314 participants (canonical benchmark closed this)
- Healthy-pool enrichment without source-conditional calibration
- Synthetic data generation without a materially better realism gate

Guardrails carried forward

- RevalExo and NONAN stay frozen — never used for model selection
- Every future experiment must pass the cross-dataset evidence-gate checklist
- No claim of clinical readiness from development-only evidence

Source: [reports/TEST_SET_CREDIBILITY_AUDIT_2026-09-03.md](#) ·

[reports/EVIDENCE_GATE_CROSS_DATASET_IMU_2026-09-03.md](#) ·

[reports/BINARY_DATA_READINESS_AND_PUBLIC_RECRUITMENT_2026-09-01.md](#)

Sprint Reproducibility Appendix

Key New Notebooks & Scripts

Key New Metrics Files

File	File	Produces	Key m
29_evidence_gated_source_only_domain_generalization.ipynb	evidence_gated_lower_back_dg_metrics.csv	Admitted ensemble	Ense AUR
30_fp_fn_and_test_set_credibility_audit.ipynb	full_expanded_lower_back_only_revalexo_metrics.csv	Confusion planning target	LB-o exter AUR
31_score_overlap_and_heterogeneous_rescue.ipynb	voisard_nonstroreshand_negative_summary.csv	Threshold sweep + MiniROCKET fusion	Pool non- stro Robu
	normalization_variant_benchmark_repeated_seeds.csv		z-sco
		MIL	AUR